Full Paper

Propellants, Explosives, Pyrotechnics

DOI: 10.1002/prep.201200215

Factors Influencing Triacetone Triperoxide (TATP) and Diacetone Diperoxide (DADP) Formation: Part 2

Jimmie C. Oxley,*[a] James L. Smith,[a] Lucus Steinkamp,[a] and Guang Zhang[a]

Abstract: A comprehensive mechanistic study regarding acetone peroxides reveals that water has a profound effect on the formation of the solid cyclic peroxides, TATP, and DADP. The identification and rate of occurrence of reaction intermediates as well as compositions of the final products offer explanation for previously reported results indicating that acid type and hydrogen peroxide concentration affect the acid catalyzed reaction between acetone and hydrogen peroxide. A kinetics study of the decomposition of TATP revealed the effects of water and alcohols. They generally retard conversion of TATP to DADP and lead to complete decomposition of TATP by acid. A mechanism is proposed for the production of TATP and DADP.

Keywords: Triacetone triperoxide (TATP) · Diacetone diperoxide (DADP) · Product formation · Product decomposition · Mechanism · Kinetics

1 Introduction

Organic peroxides are often used as polymerization catalysts or bleaching agents [1–3]. However, a few with high ratios of peroxide functionality to ketone have found use as illicit explosives [4]. We have previously reported attempts to prevent synthesis of TATP in improvised settings [5]. That work pointed out a need for a detailed mechanistic study. Some of the questions of interest in TATP formation/destruction were conditions under which TATP and DADP form, mechanisms of formation and destruction and whether DADP could be formed directly or only through a TATP intermediate. These questions were addressed by identifying intermediates by gas chromatography/mass spectrometry (GC/MS), liquid chromatography/mass spectrometry (LC/MS) at high mass resolution and $^{1}H/^{13}C$ nuclear magnetic resonance spectroscopy (NMR). Intermediates were monitored during formation and destruction experiments to elucidate mechanisms.

2 Experimental Section

2.1 Reagents and Chemicals

HPLC-grade solvents, deuterated acetone (99.8 wt-%), and acids trifluoroacetic (99 wt-%), nitric (70 wt-%), hydrochloric (37 wt-%) and sulfuric (98 wt-%) were obtained from Fisher Scientific; deuterated acetonitrile, from Cambridge Isotopes Lab. Syntheses of TATP and DADP were previously reported [5]. For synthesis of deuterium-labeled TATP d^6-acetone was used. Precipitates were filtered, rinsed with water, dried under aspiration 30 min and re-crystallized in methanol. Final products were analyzed by GC/MS, GC/uECD, ^{1}H NMR, and ^{13}C NMR spectroscopy. Anhydrous hydrogen peroxide was prepared by dissolving L-serine (20 g) in hydrogen peroxide (65 wt-%, 20 mL) [6]. To confirm its concentration, the hydrogen peroxide was dissolved in acetonitrile (ACN) and titrated with 0.25 N potassium permanaganate.

2.2 GC/MS Method

An Agilent 6890 gas chromatograph with 5973 mass selective detector (GC/MS) was used with inlet in splitleess mode at 110 °C, with purge flow 20 mL min^{-1} at 0.5 min, total helium flow of 24.1 mL min^{-1} and a Varian VF-200 MS column (15 m, 0.25 mm inner diameter, 0.25 μm film) under constant helium flow (1.5 mL min^{-1}). Oven temperature was held at 40 °C for 2 min, ramped 10 °C min^{-1} to 70 °C, then 20 °C min^{-1} to 220 °C; post-run oven was held 3 min at 310 °C. Transfer line, MS detector source, and quadrupole were held at 150 °C, 150 °C, and 106 °C, respectively. Chemical ionization used anhydrous ammonia.

2.2.1 TATP Formation (GC/MS)

All reagents were chilled to 0 °C. Hydrogen peroxide (HP) (67 wt-%, 0.47 g, 9.3 mmol) in acetonitrile (10 mL) was stirred in a round-bottomed flask, while sulfuric acid (0.95 g, 96.5 wt-%, 9.3 mmol) and then acetone (0.54 g.

[a] *J. C. Oxley, J. L. Smith, L. Steinkamp, G. Zhang*
University of Rhode Island
Chemistry Department
51 Lower College Road
Kingston, RI 02881, USA
**e-mail: joxley@chm.uri.edu*

© 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

9.3 mmol) were added dropwise. Periodically, 100 μL of the solution was removed, placed in CH_2Cl_2 (1 mL) and rinsed with 3 wt-% sodium hydrogen carbonate. Organic layer was dried with $MgSO_4$ and analyzed by GC/MS.

2.2.2 Effect of Water (GC/MS)

Acetone and HP were mixed and chilled to 0 °C. Water was mixed with sulfuric or hydrochloric acid, chilled to 0 °C, and added dropwise to the acetone/HP mixture keeping the temperature below 5 °C. The ratio of HP:acetone:acid was maintained at 1:1:1 (8.6 mmol). Once all acid was added the mixture was removed from the ice water bath and allowed to stir at room temperature 24 h; resulting products were analyzed by GC/MS.

2.2.3 TATP Destruction (GC/MS)

TATP (100 mg) was placed in a 40 mL vial and solvent (5 mL) was added. In a second 40 mL vial, sulfuric acid (200 μL, 96.5 wt-%) was added to a solvent/water mix of total volume 5 mL. Both vials were equilibrated at 45 °C. Once equilibrated, the two solutions were mixed. At intervals 100 μL aliquots were removed, quenched as above and analyzed by GC/MS. TATP and DADP were quantified by external calibration.

2.2.4 Acetone Exchange Reactions (GC/MS)

TATP was stirred at room temperature with sulfuric acid (2.6 mmol) in aqueous ethanol spiked with d^6-acetone (16.4 mmol). After 24 h the TATP had not completely dissolved, but a small aliquot of solution was analyzed by GC/MS for acetone exchange by identification of presence or absence of deuterium containing fragments. Similar experiments involving ACN, methanol, or chloroform/TFA as solvent were conducted.

2.3 LC/MS Method

Certain experiments identified intermediates by liquid chromatography/mass spectrometry at high mass resolution (LC/MS). The Thermo Scientific Exactive Orbitrap MS was operated in positive ion mode using atmospheric pressure chemical ionization (APCI). Decomposition was minimized during analysis by setting the vaporizer at 175 °C and capillary at 125 °C. Discharge current was 5 μA; the sheath gas and auxiliary gas operated at 25 and 10 arbitrary units, respectively. The LC was ramped from 70/30 methanol/ammonium acetate (4 mM) in water to 15/85 methanol/ ammonium acetate (4 mM) in 5.5 min followed by a 30 s hold. The eluent was returned to 70/30 methanol/ ammonium acetate and held for 4 min. Resolution was set to high (50,000 at 2 Hz), and the maximum injection time was 250 ms.

2.3.1 TATP Formation (LC/MS)

Acetone and HP (67 wt-%) mixtures were prepared as molar ratios of 1:1, 2:1, and 1:2 and held at room temperature without stirring. For LC/MS analysis 100 μL of each mixture was diluted to 1 mL with methanol.

2.3.2 d^6-Acetone Insertion into Proteo-TATP (LC/MS)

At room temperature, trifluoroacetic acid (TFA, 150 μL, 2.0 mmol) was added to a stirred solution of TATP (222 mg, 1.0 mmol) and d^6-acetone (75 μL, 1.0 mmol) in 10 mL ACN, methanol or chloroform was added. Every 24 h, 0.5 mL of the mixture was removed and diluted to 1.5 mL total volume for LC/MS analysis.

2.4 Nuclear Magnetic Resonance (NMR) Method

A Bruker Avance III nuclear magnetic resonance (NMR) spectrometer with 7.1 Tesla magnet was used for all NMR experiments. Kinetic experiments, at 15 °C (288 K), monitored ^1H spectra every 5 min (including scanning time) for up to 17 h or daily for up to four days. Following the kinetics experiments, the samples were neutralized with excess sodium hydrogen carbonate and either diluted by the identical solvent, sans-deuterium, and analyzed by GC/MS or returned to an NMR tube for subsequent analysis. 2-D NMR experiments: HSQC (hetero-nuclear single quantum coherence) and HMBC (heteronuclear multi-bond coherence) were performed. The ^1H (300 MHz) and ^{13}C (75 MHz) chemical shifts corresponding to all species present in the formation and decomposition of TATP reaction mixtures were obtained. NMR samples were then analyzed by GC/MS.

2.4.1 Formation of TATP/DADP (NMR)

In a 10 mL vial, HP (1 mL, 65 wt-%, 24.6 mmol) was mixed with acetone (1.9 mL, 25.8 mmol). 100 μL of the mixture were transferred into an NMR tube with CD_3CN (1.2 mL). Trifluoroacetic acid (TFA) (20–80 μL, 0.26–1.04 mmol) and trace of tetramethylsilane (TMS for calibration and quantification) were added.

2.4.2 Decomposition of TATP or DADP (NMR)

TATP (45 mg, 0.2 mmol) was dissolved in deuterated solvent (0.6 mL CD_2Cl_2, 1.0 mL $CDCl_3$ or 1.2 mL CD_3CN), transferred to 5 mm NMR tube, and acid added [TFA (20–100 μL, 0.26–1.3 mmol), sulfuric (10–20 μL, 0.18–0.36 mmol) or hydrochloric acid (10 μL, 0.12 mmol)]. DADP (29 mg, 0.2 mmol) dissolved in $CDCl_3$ (1 mL) was decomposed with TFA (60–100 μL, 0.78–1.3 mmol) or sulfuric acid (10 μL, 0.18 mmol).

© 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

These are not the final page numbers!

2.4.3 Acetone Exchange without Acid (NMR)

In 5 mm NMR tubes with 100 μL CH_2Cl_2, TATP (111 mg, 0.5 mM), and d^6-acetone (1 mL) or d^{18}-TATP (120 mg, 0.5 mM), acetone (0.9 mL), and 100 μL d^6-acetone (for lock) were sealed and stored at room temperature for a week.

2.4.4 Acetone Exchange Reactions (NMR)

To d^{18}-TATP (48 mg, 0.2 mmol) in CD_3CN (1.2 mL) or $CDCl_3$ (1.0 mL) was added 50 μL (0.68 mmol) or 40 μL (0.55 mmol) acetone, respectively. The solution was transferred to an NMR tube and TFA (20–40 μL, 0.26–0.52 mmol) was added.

2.4.5 1,3-Dichloroacetone Insertion into Proteo-TATP and d^{18}-TATP (NMR)

A tenth millimole TATP (22 mg) or d^{18}-TATP (25 mg) was dissolved in CD_3CN (1.2 mL); 1,3-dichloroacetone, 1,1-dichloroacetone (42 mg, 0.3 mmol), or monochloroacetone (27 μL, 0.3 mmol) was added, and the solution placed in a 5 mm NMR tube. TFA (40 μL, 0.54 mmol) was added.

3 Results and Discussion

3.1 Formation of TATP with Acid

Previous studies have shown that the best yield of TATP is obtained from a 1 : 1 mol ratio of acetone and hydrogen peroxide [5, 7]. When using an acid catalyst such as hydrochloric or sulfuric acid a white precipitate is quickly formed that can be washed and re-crystallized yielding high purity TATP, DADP, or a mixture of the two [5, 8]. To fully understand the mechanism of TATP and DADP formation it was necessary to conduct experiments using a co-solvent that would prevent precipitation from solution and not interfere with the analysis of the products and intermediates. Using GC/MS and NMR, TATP, DADP, and intermediate species were observed and monitored over time. Figure 1 shows the progress of a typical reaction under highly acidic conditions by monitoring TATP and DADP by GC/MS. Initially the concentration of TATP rises sharply, while that of DADP rises more gradually and levels off. The newly formed TATP undergoes decomposition in the presence of 1 molar equivalent of acid, while DADP does not. Under less acidic conditions (3.5 : 3.5 : 1 HP:acetone:sulfuric acid) TATP concentration reached a maximum, remained constant for several days, and then gradually decreased as DADP concentration continually increased. DADP is the final product if TATP cannot precipitate from solution. In agreement with NMR spectroscopic data, upon increasing the amount of acid added, a faster rise to equilibrium was observed as well as a higher equilibrium concentration of TATP in solution. The decomposition of TATP also occurred more rapidly with increased amounts of acid, and water, itself, had an effect. Figure 2 shows that when the molar ratios of acetone, hydrogen peroxide, and acid are kept constant (9.3 mmol) added water and reduced acid slows the rate of formation of both TATP and DADP and appears to suppress DADP formation more significantly. With minimal water present the rate of formation for TATP and DADP are at a maximum although TATP is still the major product observed early in the reaction.

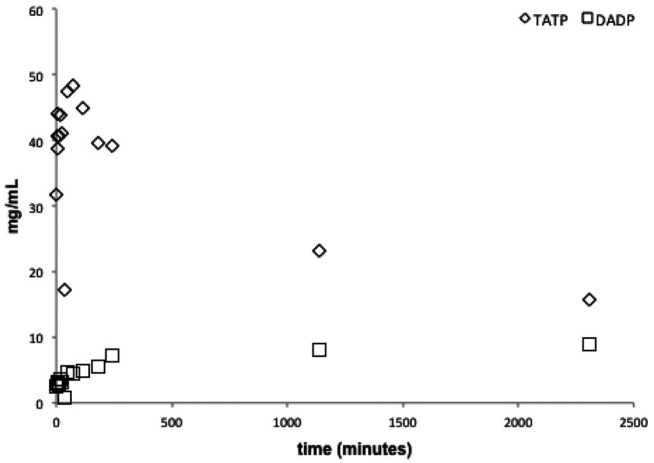


Figure 1. GC/MS data 1 : 1 : 1 (9.3 mmol) of 65 %HP to acetone to 96.5 % sulfuric acid at 0 °C.

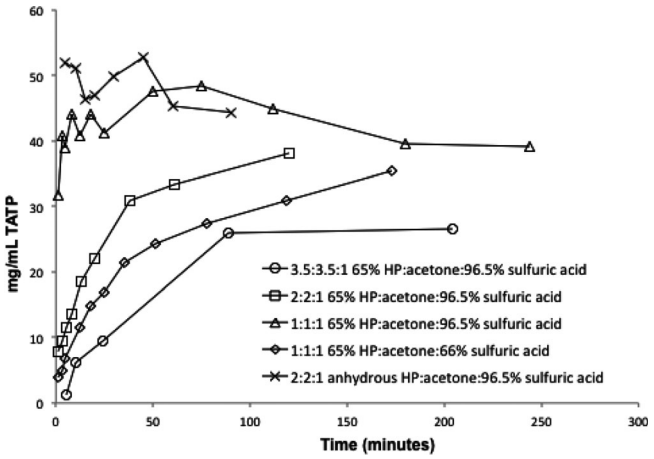


Figure 2. GC/MS data shows effect of varying the mole ratio of acid and the concentration of acid and hydrogen peroxide. Increasing the water content slows the rate of formation of TATP.

3.2 Formation of TATP without Acid

GC/MS analysis of the products when 70 wt-% hydrogen peroxide (HP) and acetone were mixed highlighted the importance of the ratios. When HP was in excess 5 : 1 over acetone more TATP was produced than DADP. When the ratio of HP to acetone was adjusted from 5 : 1 to 1 : 1 and then to 1 : 5, the total amount of solid product decreased and the amount of DADP increased relative to TATP. The reaction between HP (70 wt-%) and acetone without acid

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

Table 1. GC/MS and NMR assignments for various products of acetone/HP reaction.

	1:1 HP:acetone no acid, in acetonitrile	CI mass spectrum (NH_3)		^{1}H NMR	^{13}C NMR	^{13}C NMR	Relative NMR abundance on day							
		mass	amount	(CH_3) ppm	(CH_3) ppm	(CO) ppm	0	1	3	5	7	10	12	14
I	2,2-hydroxy hydroperoxy propane	92.1	s	1.38	24.2	102.4	L	L	M	M	S	S	S	S
II	2,2-dihydroperoxy propane	108.1	m	1.38	20.2	109.4	M	L	L	L	L	L	L	L
III	2,2'-dihydroxy-2,2'-diisopropyl peroxide	150.2	m	1.44	24.66	102.5	T	S	S	S	S	S	S	S
IV	2,2'-hydroxy hydroperoxy-2,2'-diisopropyl peroxide	166.2	--		uncertain									
V	2,2'-dihydroperoxy-2,2'-diisopropyl peroxide	182.2	l	1.44	20.5	109	T	M	M	M	L	L	L	L
	TATP (3,3,6,6,9,9-hexamethyl-1,2,4,5,7,8-hexoxonane)	222.2	vary	1.42	20.97	107.8	T	S	S	S	S	S	S	S
	DADP (3,3,6,6-tetramethyl-1,2,4,5-tetroxonane)	148.2	vary	1.35, 1.79	20.8, 21.1	108.7				T	T	T	S	S
	acetone			2.1	31	209	L	L	L	L	L	L	L	L
	dihydroxy trimer	224.3	s											
	hydroxy hydroperoxy trimer	240.3	--											
	dihydroperoxy trimer	256.3	m											

* CI with NH_3, M+1, M+18 were observed & for compound I M+35.

L = large; M = medium; S = small; T = tiny.

was monitored for up to 14 d. A number of peaks appeared in ^{1}H NMR and ^{13}C spectra as well as in GC/MS chromatogram/spectra. Assignments of intermediates by NMR and GC/MS are given in Table 1. Acetone and HP (70%) were combined in ratios 1:1, 1:10 (mostly HP), and 10:1 (excess acetone) in 0.6 mL d^3-acetonitrile and monitored by ^{1}H NMR and ^{13}C NMR for 14 d. The relative reaction progress in terms of proton resonances (as large, medium, small, or tiny peaks) is given in the right columns of Table 1. On day zero there were two prominent methyl resonances in the ^{1}H NMR (Table 1): one at 2.1 ppm, assigned to the methyl protons of acetone, and the other at $\delta = 1.38$ ppm attributed to 2-hydroxy-2-hydroperoxypropane (I) (Figure 3 shows structures of intermediates) [9]. The acetone resonance shifted to slightly higher ppm with moderate to excessive amounts of HP. This was taken as evidence for protonation of acetone by HP. By day 4 the 1:1 and 10:1 HP:acetone (i.e. moderate to large amounts of HP) samples exhibited an additional methyl resonance at $\delta = 1.44$ ppm in ^{1}H NMR, which was assigned to a dimeric species where two molecules of acetone were linked by a peroxide functionality. By that time the protonated acetone species had decreased substantially with increasing reaction intermediates.

The larger chemical shift range of the ^{13}C NMR spectrum offered better peak separation with changes in carbon functionality. On day zero for the sample with excess acetone, resonances were observed at $\delta = 24$ ppm in the methyl region and 102 ppm in the carbonyl region. These were assigned to species with hydroxy terminal groups. On day zero of the samples with excess HP, peaks were seen at $\delta = 20$ ppm in the methyl region and at $\delta = 109$ ppm in the carbonyl region. These were assigned to species with termi-

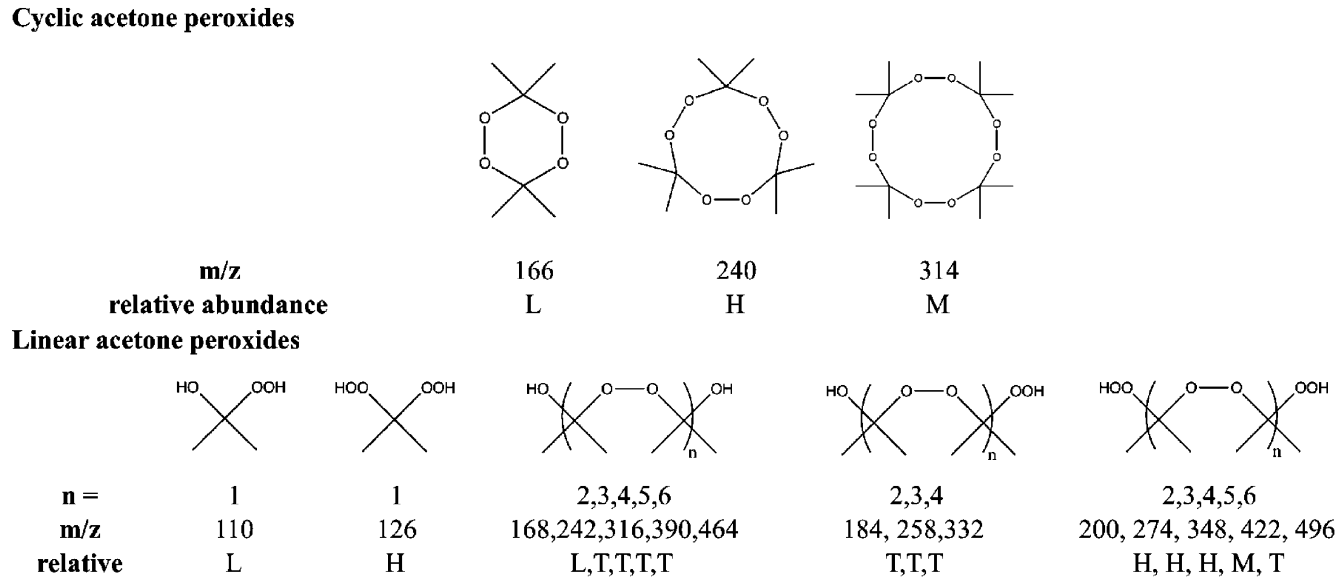


Figure 3. Products of HP/acetone reaction as observed by LC/MS.

© 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

These are not the final page numbers!

nal hydroperoxy groups. In the ^{13}C NMR spectrum of the sample of 1:1 HP:acetone, the resonance associated with the methyl groups of 2-hydroxy-2-hydroperoxypropane (I) and 2,2-dihydroperoxypropane (II) were observed at $\delta = 20$ and 24 ppm, respectively, as well as in the carbonyl region at 102 and 109, respectively. With daily monitoring, new ^{13}C resonances were observed in the methyl region: four between 20–21 ppm and three between 24–25 ppm. One at ca. 21 ppm is known to be TATP. On day zero 2-hydroxy-2-hydroperoxypropane (I) was at a maximum but diminished over time. The ^{13}C resonances of TATP ($\delta = 107.8$ ppm CO) and DADP ($\delta = 108.7$ ppm CO) did not become discernible until day 5, although their presence was detected on day 1 using GC/MS (Table 1).

In order to validate proton and carbon assignments, 2D NMR experiments were performed: HSQC (heteronuclear single quantum coherence) correlated to methyl and heteronuclear multi-bond coherence (HMBC) correlated to carbonyls. Chemical ionization GC/MS analyses, with ammonia reagent gas, were performed on the aged NMR solutions to confirm assignments of NMR resonances. Reasonable NMR resonances with corresponding masses, relative abundance of species, and chemical intuition were used to formulate assignments shown in Table 1.

We had shown that solutions of acetone/HP without added acid contained several intermediates, and LC/MS confirmed the presence of longer chain oligomers and cyclic species (Figure 3) [10]. Species with terminal peroxide functionalities are favored, and increasing amount of HP enhances their formation. Only TATP and DADP precipitated under reaction conditions, where acid was present and when no co-solvent was employed. In the absence of acid, solid TATP precipitated when the samples were aged at room temperature for up to two months. The asymmetric peroxides and longer chain oligomers were not observed by GC/MS, but using LC/MS they were observed in trace amounts.

3.3 Acetone Exchange in TATP

Experiments were conducted to determine if a new molecule of acetone could insert into an intact TATP ring. Deuterated acetone (d^6-acetone) was stirred with h^{18}-TATP and monitored daily for seven days by GC/MS. Similarly, d^{18}-TATP was monitored in h^6-acetone by ^1H NMR, typically for days, afterwards the solution was examined by GC/MS. GC/MS results indicated 1, 2, and 3 molecules of d^6-acetone were incorporated into h^{18}-TATP and 1 or 2 molecules of d^6-acetone into h^{12}-DADP in the presence of acid (Table 2). However, when no acid was added, ^1H NMR indicated no insertion of acetone; the proton resonances of the methyl groups in h^{18}-TATP ($\delta = 1.43$ ppm) showed no decrease in intensity though followed for 7 d in d^6-acetone. Likewise the resonance of h^6-acetone ($\delta = 2.1$ ppm) containing dissolved d^{18}-TATP indicated no decrease in intensity over a 7 d period. Without acid, neither exchange nor synthesis of fresh TATP nor DADP was observed.

Table 2. Acid catalyzed insertion of labeled acetone into TATP.

Labeled Acetone Insertion of TATP at 15°C for 17 hours

Experiment	Solvent	mL Solvent	Acid	µL Acid	mmols Acid	mg TATP	mmol TATP	Ratio Acid:TATP	µL Acetone	mmols acetone	Ratio acetone : TATP	Analysis	TATP	DADP
	h^6-Acetone + d^{18}-TATP 288K													
h^0-acetone + d^{18}-TATP	CD_3CN	1.2	TFA	30,40	0.40,0.54	48	0.20	2.0, 2.7	50	0.68	3.40	^1H-NMR	insertion	insertion
h^0-acetone + d^{18}-TATP	$CDCl_3$	1.0	TFA	20,30	0.27,0.40	48	0.20	1.3,2.0	40	0.54	2.72	^1H-NMR	insertion	insertion
h^0-acetone + d^{18}-TATP	d^6 & h^6 acetone	1.0	none	0	0	120	0.50	--	900	12.2	24.5	^1H-NMR LC/MS	no insertion 7 days	no insertion 7 days
	d^6-Acetone + h^{18}-TATP Reaction for 7 days 298K													
d^0-acetone + h^{18}-TATP	d^6 acetone	1.0	none	0	0	111	0.50	--	1000	13.6	27.2	^1H-NMR LC/MS	no insertion 7 days	no insertion 7 days
d^6-Acetone + h^{18}-TATP	acetone+ water or EtOH	10	sulfuric	140	2.6	222	1.0	2.6	1200	16.4	16.4	GC/MS	d^6TATP, TATP after 24 hr	d^{12} DADP, d^6 DADP & DADP+other in 48h
d^6-Acetone + h^{18}-TATP	CH_3OH, ACN, or $CHCl_3$	10	TFA	150	2.01	222	0.92	2.2	75	1.02	1.1	LC/MS	d^6, d^{12}, d^{18} TATP observed	insertion, d^{12} DADP major, d^6 DADP but no parent ion
	ChloroAcetone + d^{18} & h^{18}-TATP 288K (dichloroacetone is a solid)													
1,3 dichloro acetone + d^{18}-TATP	CD_3CN	1.2	TFA	40	0.54	22	0.10	5.4	42 mg	0.33	3.3	^1H-NMR & GC/MS	1 dichloro in TATP	no insertion
1,3 dichloro acetone + h^{18}-TATP	CD_3CN	1.2	TFA	40	0.54	22	0.10	5.4	42 mg	0.33	3.3	^1H-NMR & GC/MS	1 dichloro in TATP	no insertion
1,1 dichloro acetone + h^{18}-TATP	CD_3CN	1.2	TFA	40	0.54	22	0.10	5.4	42 mg	0.33	3.3	GC/MS	no insertion	no insertion
monochloro acetone + h^{18} TATP	CD_3CN	1.2	TFA	40	0.54	22	0.10	5.4	27	0.33	3.3	GC/MS	no insertion	no insertion
1,3 dichloro acetone + h^{18}-TATP	CD_3CN	1.2	none	0	0	22	0.10	0	42 mg	0.33	3.3	GC/MS	no insertion	no insertion

© 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

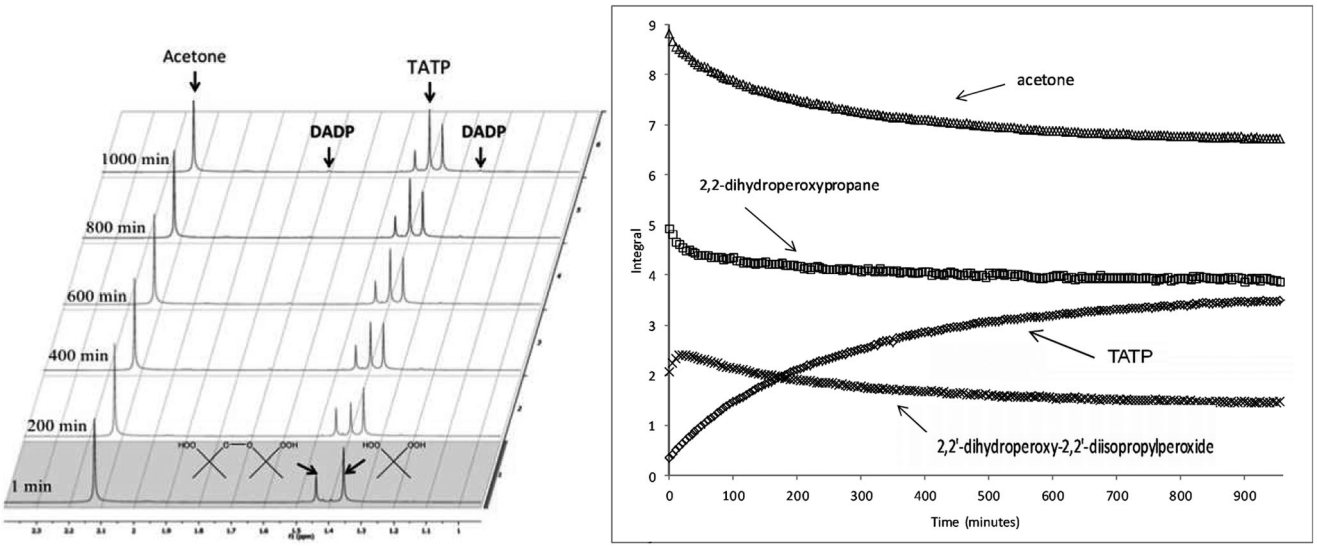


Figure 4. Formation of TATP followed by ^{1}H NMR using 40 μL TFA.

Exchange studies with chlorinated acetone were performed. Deuterated TATP or proteo-TATP was stirred with 1,3-dichloroacetone in ACN with a fivefold excess of TFA. GC/MS analysis after d^{18}-TATP and TFA had been stirred with dichloroacetone in ACN 17 hr showed d^{12}-DADP, 1,3-dichloroacetone, and dichloro-TATP. While d^{12}-DADP was formed, presumably from opening of the d^{18}-TATP-ring, no chloro-substitution into DADP was observed. Furthermore, the GC/MS fragmentation pattern as well as the quartet in the ^{1}H NMR spectrum suggested only singly substituted TATP (i.e.1,3-dichloroacetone) was formed. Neither tetrachloro-DADP nor hexachloro-TATP was observed, suggesting their formation may be sterically hindered. Interestingly, no incorporation of 1,1-dichloroacetone nor monochloroacetone was observed (^{1}H, ^{13}C, GC/MS) even in the presence of acid (Table 2). Nevertheless, the observation of dichloro-substituted TATP indicates ring opening and re-closing does occur.

3.4 Rate of TATP Formation

^{1}H NMR was used to monitor the one-to-one reaction of acetone and HP (25 mmol each) at 15 °C using 20, 40 or 80 μL of trifluoroacetic acid (TFA) (0.27, 0.54, 1.07 mmol). The reaction was monitored by taking spectra every 5 min (including scan time) for up to 17 h (1020 min) (Figure 4 shows typical NMR spectroscopic data). After 200 min the methyl protons of TATP were clearly visible (as opposed to three days without acid). DADP protons only were barely visible after 800 min and were still faint at $\delta = 1.35$ and 1.79 ppm in the 1020 min spectrum. The most abundant intermediates in the acid catalyzed reactions observed by NMR and GC/MS were 2,2-dihydroperoxypropane (II) and 2,2′-dihydroperoxy-2,2′-diisopropylperoxide (V). Although 2-hydroxy-2-hydroperoxypropane (I) was most abundant when there was no acid catalyst, formation of 2,2-dihydroperoxypropane (II) was favored under acidic conditions [7, 11]. TATP formation was greatly accelerated by addition of acid; yet acid also caused TATP decomposition, as evident from following ^{1}H NMR resonance of TATP ($\delta = 1.42$ ppm) when treated with various volumes of TFA (Figure 5). Monitoring TATP formation reactions by GC/MS showed the effect of the strength of acid as well as type of acid used. Compared to a mixture of acetone and HP with no added acid, concentrated sulfuric acid greatly enhanced the rate of formation of TATP as well as its decomposition to DADP. Reducing the concentration of the sulfuric acid, while maintaining a 1 molar equivalent acid, slowed TATP formation and significantly inhibited its decomposition to

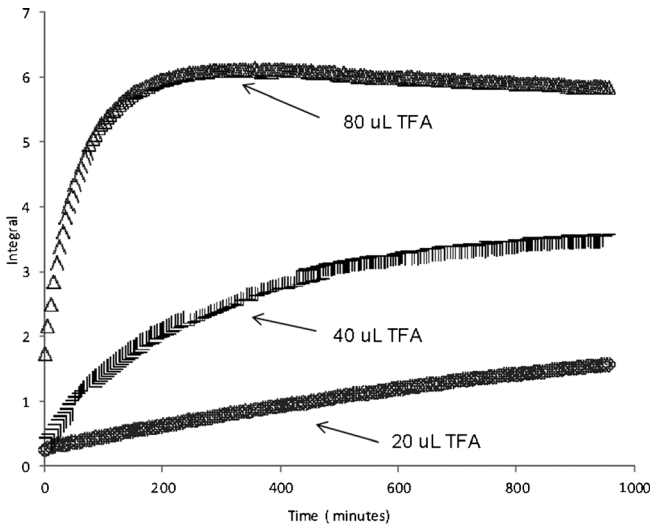


Figure 5. Formation of TATP followed by ^{1}H NMR 20, 40, and 80 μL TFA.

© 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

These are not the final page numbers!

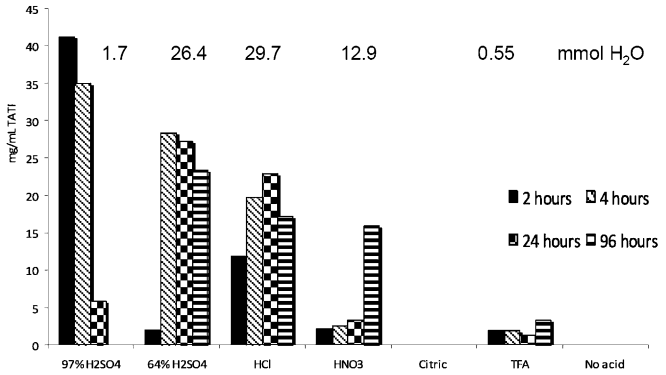


Figure 6. Rate of TATP formation with 8.6 mmol each HP-acetone-acid versus water.

DADP. This observation suggested that water played a role. Concentrated HCl contains significantly more water than concentrated H_2SO_4. When water was added to sulfuric acid so that the water content in the acetone/HP mix was the same as when HCl was used, the rates of TATP formation with either acid were comparable. Use of nitric acid did not result in the same observed rate of formation, but nitric acid is the weakest of the strong acids used in these experiments. Use of weak acids such as citric acid and TFA resulted in dramatically slower rates of TATP formation (Figure 6), although TFA was still capable of decomposing TATP.

3.5 Effect of Water

Previously reported results indicate that a change in the concentration of acid and hydrogen peroxide can dramatically affect the outcome of TATP syntheses [5]. Dilute reagents result in poor yield of solid products, and concentrated reagents in the presence of higher acid loadings result in increased DADP formation. Water appears to play an important role in the synthesis. In an attempt to understand how water affects DADP vs. TATP formation, several syntheses were attempted using 30 wt-% and 50 wt-% HP, acetone and concentrated sulfuric or hydrochloric acid. The ratios of the three reagents were maintained at 1:1:1 (8.6 mmol scale), but excess water, over that contributed by the reagents, was added (Table 3, Figure 6). At the lowest levels of added water, the white solid formed was 100% DADP. At the highest levels of added water, the white solid precipitating was 100% TATP [12]. When the acid added was HCl, only a small amount of DADP was observed; this is attributed to the large amount of water (63 wt-%) in HCl. This phenomenon can be explained by the tendency of 2-hydroxy-2-hydroperoxypropane to disproportionate to 2,2-dihydroperoxypropane (II) and acetone in aqueous media [9]. The formation of the dihydroperoxy species appears to be a key step in TATP formation.

Table 3. Acetone, HP, acid, 1:1:1 (8.6 mmol each), and varied H_2O.

Acetone (0.5 g), HP, Acid (1:1:1) (8.6 mmol each)				
water (mL)	yield (mg)	% yield	% TATP	% DADP
50% HP (0.586 g), 96.5% H2SO4 (0.876g)				
0.25	308	48.3	0	100
0.5	246	38.6	7.6	92
0.75	348	54.6	81	19
1	378	59.3	100	0
30% HP (0.976 g), 96.5% H2SO4 (0.876g)				
	218	34	11	89
0.25	140	22	72	28
0.5	265	42	72	28
0.75	161	25	89	11
30% HP (0.976 g), 37% HCl (0.85g)				
	92.1	14	81	0
0.25	98.7	15	83	0
0.5	92.2	14	87	0
0.75	99.3	16	94	0

3.6 Effect of Solvent and Temperature

To probe whether order of reactant addition had an effect on formation of TATP versus DADP, it was varied (Table 4). The final precipitates, as well as in-situ products, were monitored by extracting aliquots of the reaction at intervals during reagent addition and analyzing by GC/MS. There were no notable differences in the final products obtained regardless of whether the acid was added first to the acetone, to the HP, or to both together (cf. exp. 1 to 3, Table 4) [8]. Only when acid was the last reagent added did the reaction proceed slow enough for an intermediate, 2,2′-dihydroperoxy-2,2′-diisopropylperoxide (V) to be observed along with both cyclic peroxides (exp. 3); otherwise, only the cyclic peroxide(s) were seen. The final product of the hydrogen peroxide/acetone mixture varied markedly with solvent; acetonitrile favored formation of DADP at 0 °C, while alcohols favored TATP (cf. exp.1 with 1*, Table 4). The effect of temperature has been discussed by others without agreement [8, 11, 13]. Generally, TATP is favored at lower temperatures, but the effect of temperature can be manipulated by other factors such as solvent. While DADP was the major solid product in ACN at 0 °C, at lower temperatures even in ACN, TATP was favored (cf. 1 to 1′ and 3 to 3′, Table 4). When TFA was substituted for sulfuric acid (exp 1′′) no precipitate formed within the same time interval as previous reactions (ca. 30 min).

3.7 Rate of TATP Decomposition

The decomposition of TATP by acid in CD_3CN or $CDCl_3$ was monitored by ^1H NMR spectroscopy. In ACN and $CDCl_3$ decomposition of TATP was pseudo-first order and formed DADP and acetone (Table 5). Quantifying TATP formation kinetics was more difficult due to formation of a number of

© 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

Table 4. GC/MS analysis of cyclic peroxide products versus HP/acetone reaction conditions.

exp	Order of Reactant Addition HP (66.8%)	acetone	sulfuric acid (96.5%)	Conditions solvent	temp	crude yield	in situ products	% Final Product TATP	DADP	analysis GC
(repeat)	2.7g,54mmol	3.0g, 51mmol	10.4g, 102mmol	ACN (mL)	°C	gram				
1 (3)	1	3	2	30	0	2.6	TATP/DADP	--	98	uECD
2 (2)	3	1	2	30	0	3.1	DADP	--	100	uECD
3 (1)	2	1	3	30	0	3.1	TATP/dimer/DADP	--	87	MS
2'	3	1	2	30	-20	3.0	TATP/DADP	7	77	MS
1'	1	3	2	30	-25	3.3	TATP/dimer/DADP	67	22	MS
	1.5g,26mmol	1.5g, 26mmol	5.24g, 51mmol							
3'	2	1	3	15	-40	1.7	TATP/dimer/DADP	93	1	MS
1*	1	3	2	**15 EtOH**	0	1.0	n/a	98	--	uECD
1*	1	3	2	**15 MeOH**	0	0.8	n/a	100	--	MS
1'	1	3	2	15	**-40**	1.8	TATP/DADP	90	1	MS
1"	1	3	2 TFA (99%,5.8g)	15 ACN	0		dimer->TATP, DADP		no ppt	

Table 5. First-order rate constants obtained from ^{1}H NMR at 288 K following destruction of 45 mg (0.203 mmol) TATP with acid or TATP formation from 100 μL acetone and HP (1.36 mmol; 1 : 1 ratio) and TFA in 1.2 mL CD_3CN.

TATP Formation & Destruction Monitored by ^{1}H NMR

Acid	uL	mmol Acid	Rate Constant (s^{-1})
Decomposition of TATP (45mg, 0.2 mmol) in Chloroform			
TFA	30	0.392	1.50E-05
TFA	40	0.522	1.83E-05
TFA	60	0.784	6.67E-05
TFA	80	1.045	2.48E-04
SA	10	0.188	2.17E-05
SA	15	0.281	3.00E-05
SA	20	0.375	6.17E-05
Decomposition of TATP (45 mg, 0.2mmol) in Acetonitrile			
TFA	60	0.784	1.17E-05
TFA	100	1.306	3.50E-05
Decomposition of DADP (29 mg) in Chloroform			
SA	10	0.188	3.67E-05
Formation of TATP (45mg, 0.2mmol) in Acetonitrile			
TFA	20	0.261	1.75E-04
TFA	40	0.522	4.83E-04
TFA	80	1.045	4.83E-04

intermediates, but initial formation rates of TATP in ACN are estimated in Table 5. Data suggests that at very high concentrations of acid, TATP is destroyed as fast as it is formed, and destruction of TATP in ACN leads to the formation of DADP (Table 4 and Table 6). As TATP decomposed to DADP and acetone, other intermediates could be observed by ^{1}H NMR in d^3-ACN. In ACN, 2,2′-dihydroperoxy-2,2′-diisopropylperoxide (V) and 2,2-dihydroperoxypropane (II) were observed as intermediates, whereas in $CDCl_3$ only 2,2′-dihydroperoxy-2,2′-diisoproylperoxide (V) was observed. NMR and MS data suggests these intermediates are identical to those observed in formation experiments (Table 1).

Table 6 emphasizes the effect of water and solvent on TATP decomposition. Not only did water slow the decomposition, but it also appeared to retard conversion of TATP

Table 6. GC/MS monitoring of TATP (100 mg, 045 mol) + H_2SO_4 (96.5 wt-%, 200 μL, 3.6 mmol).

GC/MS monitoring of TATP Decomposition (298K)

TATP (100 mg, 045 mmol) + H2SO4 (96.5wt%, 200 uL, 3.6 mmo

Solvent (10 mL)	k (sec^{-1})	Result
Acetonitrile	2.41E-03	DADP
90:10 acetonitrile:water	1.81E-04	no DADP
80:20 acetonitrile:water	4.74E-05	no DADP
70:30 acetonitrile:water	1.82E-05	no DADP
Methanol	1.98E-03	no DADP
90:10 methnaol:water	2.13E-04	no DADP
80:20 methanol:water	2.96E-04	no DADP
Ethanol	6.43E-04	no DADP
90:10 ethanol:water	8.63E-05	no DADP
80:20 ethanol:water	1.46E-04	no DADP
70:30 ethanol:water	2.00E-04	no DADP
n-propanol	2.96E-04	no DADP
90:10 n-propanol:water	1.72E-05	no DADP
isopropanol	8.86E-05	no DADP
90:10 isopropanol:water	2.88E-05	no DADP
80:20 isopropanol:water	1.01E-04	no DADP
70:30 isopropanol:water	6.96E-05	no DADP
t-butanol	n/a	DADP/TATP
90:10 t-butanol:water	n/a	DADP/TATP

*Reaction in *tert*-butanol quickly converted some of the TATP to DADP but further change in concentrations was not observed. **Reaction in neat ACN used only 20 μL acid.

© 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

These are not the final page numbers!

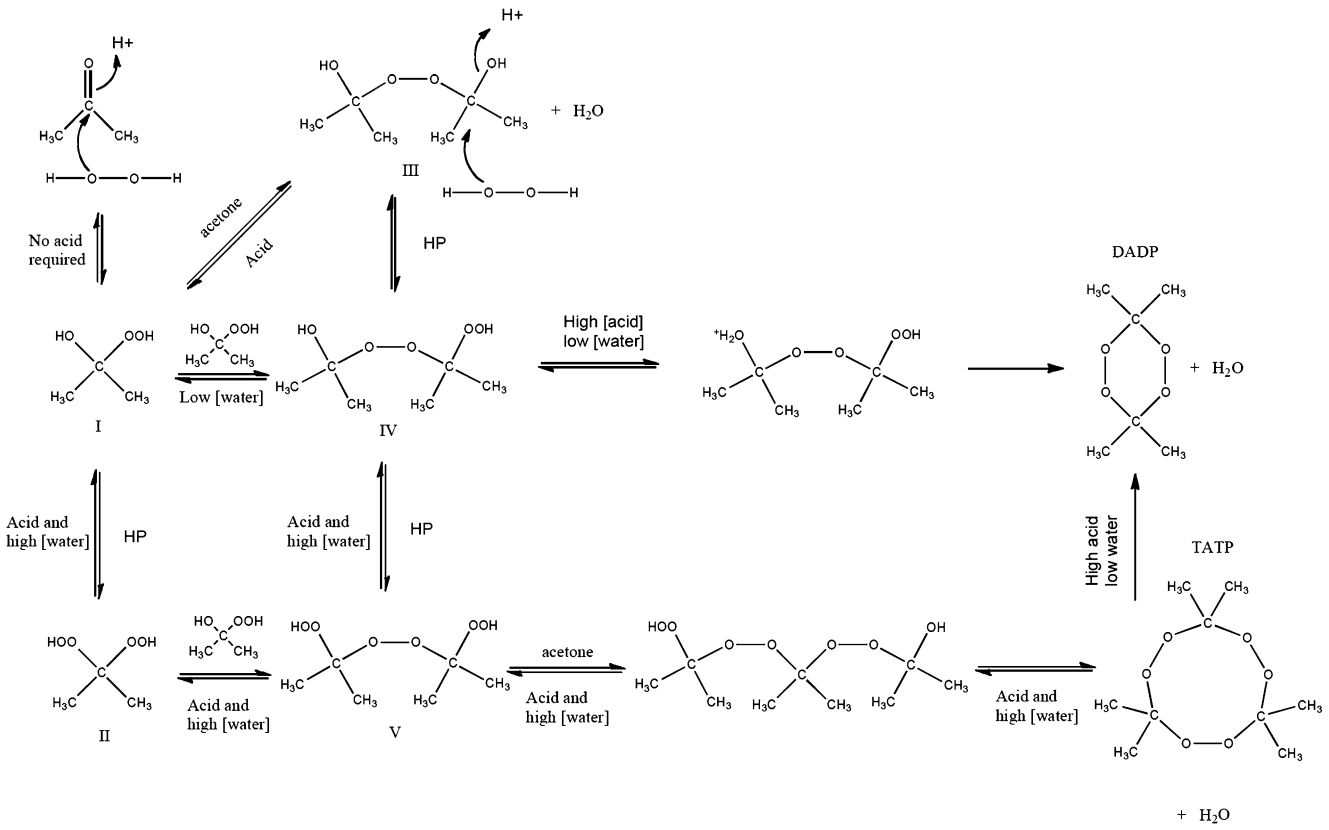


Figure 7. Mechanism for the synthesis of TATP and DADP.

to DADP. Table 6 shows the pseudo first-order rate constants for TATP destruction in ACN and various alcohols. When concentrated sulfuric acid was added to a TATP ACN solution, the rate of disappearance of TATP was very high, and DADP was the end product. With added water the reaction was slowed significantly, and only trace amounts of DADP were observed. When concentrated sulfuric acid was added to a TATP alcohol solution, the destruction of TATP was slower than in neat ACN, and only trace amounts of DADP were observed. Destruction of TATP was slower in alcohols with greater reactivity towards concentrated sulfuric acid, i.e. 2-propanol or *tert*-butanol. Methanol showed the highest rate of TATP destruction followed by ethanol, *n*-propanol and 2-propanol. When *tert*-butanol was used, an anomalous effect was observed. The reaction proceeded very quickly to a mixture of TATP and DADP and ceased. The decomposition of TATP by 37 wt-% HCl generated some DADP, but the amount was very small compared with other acids (TFA SA) and chlorinated acetone species were detected.

3.8 Mechanism

A proposed mechanism for the production of TATP and DADP is given in Figure 7. In a reaction between HP and acetone without acid catalyst, 2-hydroxy-2-hydroperoxypropane (I) was observed in high quantities soon after mixing. When acid was added, 2,2-dihydroperoxy-propane (II) was the primary species observed shortly after mixing [11]. Symmetric species, where two acetones are linked by a peroxide linkage, 2,2′-dihydroxy-2,2′-diisopropylperoxide (III) and 2,2′-dihydroperoxy-2,2′-diisopropylperoxide (V) were also observed, but at later reaction times. The asymmetric species similar to 2-hydroxy-2′-hydroperoxy-2,2′-diisopropylperoxide (IV) were not directly observed by NMR or GC/MS and only trace amounts were observed using LC/MS. We speculate that when these are formed the hydroxyl group exchanges with a hydroperoxy group, or they rapidly convert to DADP and TATP, respectively [7]. The effect of water is apparent at this point. When water content is low, 2-hydroxy-2-hydroperoxypropane (I) can be protonated facilitating formation of (IV) and a pathway to cyclization forming DADP. When water content is high, disproportionation of (I) becomes favored resulting in the formation of the dihydroperoxy species (II) and, ultimately, the formation of TATP [9]. Under high water conditions, water, itself, is protonated; and the overall reaction proceeds more slowly. Alcohol solvents also become involved in this competition for protonation. Formation reactions performed in methanol or ethanol under highly acidic conditions produced almost 100% TATP compared to the same reactions performed in acetonitrile, which produced 90 to 98% DADP (Table 4).

© 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

Destruction of TATP results in the formation of 2,2′-dihydroperoxy-2,2′-diisopropyl peroxide (V) and acetone [14]. Depending upon the solvent used 2,2-dihydroperoxypropane (II) may also be observed. Under destruction conditions, the presence of water or alcohol retards TATP loss and prevents the formation of DADP. The retarding effect of water and alcohol can be attributed to the intermediate species competing with the solvent for protonation. The lack of DADP formation can be attributed to the solubility of the intermediates in more polar and protic solvents. Under mildly acidic conditions the opening of the TATP ring is reversible and incorporation of a new acetone molecule is possible, or decomposition to DADP may occur, explaining the presence of deuterated DADP in addition to deuterated TATP.

4 Conclusions

We have reported that the synthesis of TATP is achieved in best yield by use of a 1 to 1 molar ratio of HP to acetone with modest amounts of acid (10–50 mol%). However, acid catalyzes TATP synthesis and decomposition, especially at high acid levels. Herein we examine the intermediates of the acetone/HP reaction and the rates of formation and destruction of TATP and postulate a mechanistic pathway. The oxidation of acetone by HP occurs whether or not acid is added; it is dramatically slower without acid, taking weeks and months to precipitate TATP and DADP [15]. Other linear and even cyclic peroxides were observed in the reaction solution, but only TATP and DADP precipitate out due to their low solubility in aqueous media. If an organic solvent is present, TATP does not precipitate out of solution and, in the presence of acid, converts to DADP. Whether or not acid is present as soon as HP and acetone are mixed 2-hydroxy-2-hydroperoxypropane (I) forms. This then proceeds to form 2-hydroxy-2′-hydroperoxy-2,2′-diisopropylperoxide (IV) (the precursor to DADP) or 2,2′-dihydroperoxy-2,2′-diisopropylperoxide (V) (the precursor to TATP) depending on whether water content in the reaction mix is low or high, respectively. Water, which comes into the reaction via the acid and the HP, slows the formation of TATP and DADP, DADP most dramatically. Indeed, the same effect is observed when alcohol, which like water is susceptible to protonation, is added to the reaction mixture.

Under nearly anhydrous conditions sulfuric acid and trifluoroacetic acid decompose TATP in a pseudo first-order fashion to DADP. Under mildly acidic conditions the TATP ring opens to form 2,2′-dihydroperoxy-2,2′-diisopropylperoxide (V) and acetone; this may re-cyclize with a new acetone molecule or further decompose to DADP and other smaller molecules. This point is important for the formation and destruction of TATP. Under destruction conditions it confirms that the process does not proceed to any appreciable extent via radical intermediates. The presence of the dihydroperoxy intermediates observed during destruction or under milder insertion conditions can only be rationalized by an ionic mechanism where acetone is removed leaving hydroperoxy species [14]. The presence of completely substituted TATP also implies one of two things: the TATP molecule opens and equilibrium with the intermediates is reestablished or the ring continues to open and reclose allowing all acetone molecules to be replaced with their deuterated counterparts. Both these possibilities seem reasonable; which dominates may depend upon the solvent. Regarding formation, the insertion of acetone seems to indicate that (V) reacts with acetone to form the asymmetric hydroxy hydroperoxy trimeric species, prior to cyclizing to TATP. This supports the claim that the asymmetric species are short-lived intermediates leading to the cyclic peroxide products.

References

[1] S. R. Graham, R. Hodgson, L. Vechot, M. I. Essa, Calorimetric Studies on The Thermal Stability of Methyl Ethyl Ketone Peroxides (MEKP) Formulations, *Proc. Safety Environ. Protect.* **2011**, *89*, 424–433.

[2] Y. F. Lin, J. M. Tseng, T. C. Wu, C. M. Shu, Effects of Acetone on Methyl Ethyl Ketone Peroxide Runaway Reaction, *J. Hazard. Mater.* **2008**, *153*, 1071–1077.

[3] X. Jia, Y. Wu, P. Liu, Effects of Flour Bleaching Agent on Mice Liver Antioxidant Status and ATPases, *Environ. Toxicol. Pharmacol.* **2011**, *31*, 479–484.

[4] *Encyclopedia of Explosives and Related Items*, Vol. 1 (Ed.: B. T. Fedoroff), Picatinny Arsenal, NJ 1960 A41 and Vol. 7 (Ed.: S. M. Kaye), Picatinny Arsenal, NJ 1975 H79.

[5] J. C. Oxley, J. L. Smith, P. Bowden, R. Rettinger, Factors Influencing TATP and DADP Formation: Part I, *Propellants Explos. Pyrotech.* submitted.

[6] Y. Wolanov, L. Ovadia, A. V. Churakov, A. G. Medvedev, V. M. Novotortsev, P. V. Prikhodchenko, Preparation of Pure Hydrogen Peroxide and Anhydrous Peroxide Solutions From Crystalline Serine Perhydrate, *Tetrahedron* **2010**, *66*, 5130 – 5133.

[7] K. J. McCullough, Ketone-Derived Peroxides. Part 1. Synthetic Methods, *J. Chem. Res. Synopses* **1980**, *2*, 601–627.

[8] R. Matyas, J. Pachman, Study of TATP: Influence of Reaction Conditions on Product Composition, *Propellants Explos. Pyrotech.* **2010**, *35*, 31–37.

[9] M. C. V. Sauer, J. O. Edwards, Reactions of Acetone and Hydrogen Peroxide. I. The Primary Adduct, *J. Phys. Chem.* **1971**, *75*, 3004 – 3011; M. C. V. Sauer, J. O. Edwards, Reactions of Acetone and Hydrogen Peroxide. II. Higher Adducts, *J. Phys. Chem.* **1972**, *76*, 1283 – 1287.

[10] M. E. Sigman, C. D. Clark, T. Caiano, R. Mullen, Analysis of Triacetone Triperoxide (TATP) and TATP Synthetic Intermediates By Electrospray Ionization Mass Spectrometry, *Rapid Commun. Mass Spectrom.* **2008**, *22*, 84 – 90.

[11] P. R. Story, B. Lee, C. E. Bishop, D. D. Denson, P. Busch, Macrocyclic Synthesis. II. Cyclohexanone Peroxides, *J. Org. Chem.* **1970**, *35*, 3059 – 3062.

[12] J. R. Sanderson, A. G. Zeiler, R. J. Wilterdink, An Improved Synthesis of Dicyclohexylidene Diperoxide, *J. Org. Chem.* **1975**, *40*, 2239 – 2241.

[13] A. J. Pena, L. P. Pacheco-Londoño, J. Figueroa, L. A. Rivera-Montalvo, F. R. Román-Velazquez, S. P. Hernández-Rivera, Characterization and Differentiation of High Energy Cyclic Organic

 © 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

Peroxides by GC/FT-IR, GC-MS, FT-IR and Raman Microscopy, Sensors, and Command, Control, Communications, and Intelligence (C3I) Technologies for Homeland Security and Homeland Defense IV, *SPIE Proceedings*, *5778*, 347–358.

[14] D. Armitt, P. Zimmermann, S. Ellis-Steinborn, Gas Chromotography/Mass Spectrometry Analysis of Triacetone Triperoxide (TATP) Degradation Products, *Rapid Commun. Mass Spectrom.* **2008**, *22*, 950 – 958.

[15] S. A. Wilson, J. E. Brady, J. L. Smith, J. C Oxley, The Risk of Mixing Dilute Hydrogen Peroxide and Acetone Solutions, *J. Chem. Health Safety* **2012**, *19*, 27 – 33.

Received: December 21, 2012
Revised: March 14, 2013
Published online: ■■ ▮▮, 0000

© 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim